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from anthropic import Anthropic
from dotenv import load_dotenv
import os
import subprocess

load_dotenv(override=True)
client = Anthropic(base_url=os.getenv("ANTHROPIC_BASE_URL"))
MODEL = os.getenv("MODEL_ID")

SYSTEM = f"You are a coding agent at {os.getcwd()}. Use bash to solve tasks. Act, don't explain."

TOOLS = [{
    "name": "bash",
    "description": "Run a shell command.",
    "input_schema": {
        "type": "object",
        "properties": {"command": {"type": "string"}},
        "required": ["command"],
    },
}]

def run_bash(command: str) -> str:
    dangerous = ["rm -rf /", "sudo", "shutdown", "reboot", "> /dev/"]
    if any(d in command for d in dangerous):
        return "Error: Dangerous command blocked"
    try:
        r = subprocess.run(command, shell=True, cwd=os.getcwd(),
                           capture_output=True, text=True, errors="replace",
                           timeout=120)
        out = (r.stdout + r.stderr).strip()
        return out[:50000] if out else "(no output)"
    except subprocess.TimeoutExpired:
        return "Error: Timeout (120s)"
    except (FileNotFoundError, OSError) as e:
        return f"Error: {e}"

def agent_loop(messages: list):
    while True:
        # 1. 把整本账本发给模型
        response = client.messages.create(
            model=MODEL, system=SYSTEM, messages=messages,
            tools=TOOLS, max_tokens=8000,
        )

        # 2. 模型的回复记进账本
        messages.append({"role": "assistant", "content": response.content})

        # 3. 这轮有没有调工具? 没有 -> 任务完成, 退出
        tool_calls = [
            block for block in response.content if block.type == "tool_use"
        ]

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    if not tool_calls:
        return

    # 4. 有 → 逐个执行，收集结果
    results = []
    for block in tool_calls:
        print(f"$ {block.input['command']}")
        output = run_bash(block.input["command"])
        print(output[:200])
        results.append({
            "type": "tool_result",
            "tool_use_id": block.id,
            "content": output,
        })

    # 5. 结果塞回账本，回到第 1 步
    messages.append({"role": "user", "content": results})

if __name__ == "__main__":
    print("s01: Agent Loop")
    print("Enter a question, press Enter to send. Type q to quit.\n")

    history = []
    while True:
        try:
            query = input("agent >> ")
        except (EOFError, KeyboardInterrupt):
            break
        if query.strip().lower() in ("q", "exit", ""):
            break
        history.append({"role": "user", "content": query})
        agent_loop(history)
        # 打印模型的最终文字回复
        for block in history[-1]["content"]:
            if getattr(block, "type", None) == "text":
                print(block.text)
    print()

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